Building Agents: The Four Engineering Layers

Scaler School of Technology | ML System Design and LLMOps | Week 4 — Bonus Session | Pre-Lecture Prep

What Is This Session?

This is a bonus, optional, unassessed session. We take one AI agent — an automated pull-request reviewer — and improve it three times, pairing engineering disciplines the way you'd actually reach for them (prompt with context, loop with harness), watching the metrics move. You'll see how the same core agent can be tuned radically differently depending on which layer you invest in.

Prerequisite: What Makes an Agent?

Agent Formula

From Week 4 Session 2: *agent* = LLM + tools + memory + loop. The LLM proposes actions, tools execute them, memory carries state, the loop repeats until done.

The Four Engineering Layers

The prompt is the sentence. Context is the page. The loop is the paragraph structure. The harness is the print shop. They compose; they do not substitute.

Layer	The question it answers	You are engineering...
Prompt engineering	What do we *say* on a single call?	instructions, role, output contract, few-shot, "think before you act"
Context engineering	What *information* is in the window — and what's left out?	retrieval, compaction, ordering, trimming tool results, memory, the token budget
Loop engineering	How does one output *drive the next call*?	stop conditions, step caps, stall detection, re-planning, carried state
Harness engineering	What *system* does the loop run inside?	tool sandboxing, verification gates, cost ceilings, human-in-the-loop, observability

Setup You Must Do Before Class

- Get a free Gemini API key at `aistudio.google.com/apikey`
- `pip install google-genai matplotlib`
- Clone or pull the course repo; the demo lives at `course_content/week_4_session_3/w4s3_demo_agent_layers.ipynb`
- *Note:* the notebook has `USE_RECORDED = True` and replays cached model responses, so it runs with no key and no network — the key is only needed if you want to re-record. Setup is not a blocker.

Three Warm-Up Questions

Discuss these in class — no answers here:

1. Your agent returns answers in the wrong format half the time. Which of the four layers is the fix?
2. Your agent misses bugs that are obvious from the function *above* the changed lines. Which layer?
3. Your agent sometimes runs 30 steps on a trivial task. Which layer?

What to Bring

A laptop that can run the notebook; curiosity; the willingness to argue with the instructor.

Duration: ~1.5 hours **Prerequisites:** Week 4 Session 2 (Agents, Tool Use) **Instructor:** Priyansh Saxena